CERN ZENODO DOI: 10.5281/zenodo.23436152 · APACHE 2.0

THE MARMA-CRYPTO PROTOCOL

Cryptographic-Geometric Watermarking & Sovereign Document Sealing via Sri Yantra Nonlinear Planimetry and Elliptic Curves over $\mathbb{F}_{2^{257}}$

Ukrainian National Standard DSTU 4145-2002 · ETSI EN 319 142-1 (PAdES-B-LTA) · RFC 3161 TSP



GEOMETRIC INVARIANT

54 Triple Marmas

9 interlocking triangles form 43 sub-triangles & 54 triple intersections solved to IEEE-754 machine precision (6.13×10^{-11} rad).

SUB-MICRON BINDING

24-Bit Quantized δ_k

Maps a 512-bit DSTU 4145-2002 signature (r, s) into deterministic shifts $|\delta_k| \leq 10^{-5}$ with exact bilateral symmetry $\Delta_{\text{sym}} = 0$.

ZERO-TRUST VERIFIER

100% Client-Side

Instant verification of signed `.pdf`, `.docx`, and `.svg` files in your browser or via Debian/Ubuntu `apt install`.

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SWIPE TO EXPLORE →

Why Conventional Watermarks & Detached Signatures Fail

In high-stakes litigation, M&A, and cross-border finance, visual appearance and cryptographic metadata are completely disconnected.

FAILURE MODE A · VISUAL LAYER

Decorative Stamps & Raster Watermarks

- **Zero Cryptographic Binding:** A standard PDF stamp or QR box has no mathematical link to the underlying contract text.
- **Trivial Cloning & Stripping:** Any vector editor can copy a visual stamp onto a forged page or delete a watermark layer in seconds.
- **Raster Degradation:** Pixel-based steganography breaks under vector re-rendering, PDF/A conversion, or scaling.

FAILURE MODE B · METADATA LAYER

Detached CAdES / Hidden ASN.1 Streams

- **Invisible to Human Inspection:** The cryptographic signature lives in hidden binary `/Contents` or external `.p7s` files.
- **Metadata Stripping:** Exporting to `.docx`, `.svg`, or re-saving strips detached containers while leaving the visual page unchanged.
- **Vendor Lock-In:** Requires proprietary national PKI desktop utilities unavailable on headless Linux servers.

THE MARMA-CRYPTO BREAKTHROUGH · DUAL-LAYER CONVERGENCE

Unifying Sacred Nonlinear Geometry with National Elliptic Curve Cryptography

What if the **visible vector geometry on the page** is itself an overdetermined nonlinear mathematical system whose exact vertex coordinates encode the document’s cryptographic signature?

1. VISIBLE VECTOR SEAL

Survives across `.pdf`, `.docx`, and `.svg` vector representations.

2. SUB-MICRON RIGIDITY

Any 10^{-3} coordinate shift breaks 54 triple intersections.

3. STATUTORY QES + TSP

Full PAdES-B-LTA + RFC 3161 TSP court-admissible validity.

Nonlinear Planimetry of the 9-Triangle Sri Yantra

Why the canonical Nava-Cakra is one of the hardest overdetermined problems in computational planimetry.

THEOREM 1 · EUCLIDEAN IMPOSSIBILITY & NONLINEAR CONCURRENCY

9 Intersecting Triangles → 43 Sub-Triangles → 54 Triple Marmas

The central **Nava-Cakra** consists of **4 upward-pointing Śiva triangles** ($\Delta_{U1} \dots \Delta_{U4}$) and **5 downward-pointing Śakti triangles** ($\nabla_{D1} \dots \nabla_{D5}$) inscribed within a canonical circle of radius $R = 100$ centered at $(C_x, C_y) = (150, 150)$.

$$\det \begin{pmatrix} A_{i,1} & B_{i,1} & C_{i,1} \\ A_{i,2} & B_{i,2} & C_{i,2} \\ A_{i,3} & B_{i,3} & C_{i,3} \end{pmatrix} = 0, \quad \forall i \in \{1, ..., 54\}, \quad \Delta E \approx 6.13 \times 10^{-11} \text{ rad} < \varepsilon_{\text{float64}}$$

CANONICAL BASELINE COORDINATES (300 × 300 COORDINATE SPACE)

Exact Bilateral Symmetry Invariant: lx_k + rx_k = 300.000000000

Layer	Orientation	lx_k (Left)	rx_k (Right)	by_k (Base)	ay_k (Apex)
D1	Downward (∇)	53.65669560	246.34330440	123.20508076	250.00000000
U1	Upward (Δ)	52.98401103	247.01598897	174.24660561	50.00000000
U3	Upward (Δ)	98.71823313	201.28176687	220.03828947	123.20508076
U2	Upward (Δ)	78.26467998	221.73532002	197.92315674	78.10499178
D3	Downward (∇)	90.48569229	209.51430771	78.10499178	160.66014976
D2	Downward (∇)	80.98384839	219.01615161	103.12199145	220.03828947
U4	Upward (Δ)	114.94885006	185.05114994	160.66014976	103.12199145
D4	Downward (∇)	116.35142605	183.64857395	134.30757627	197.92315674
D5	Downward (∇)	124.61190803	175.38809197	144.79777263	174.24660561

Cryptographic Binding via National Standard DSTU 4145-2002

From normalized document text to a deterministic elliptic curve signature vector (r, s) over characteristic-two fields.

STEP 01 · CANONICAL TEXT NORMALIZATION & TRIPLE CRYPTOGRAPHIC DIGEST

Multi-Standard Hash Commitment (GOST + Kupyna-256 + SHA-256)

Document text M is extracted in deterministic page/paragraph order, stripped of dynamic seal headers, normalized via Unicode NFC, and hashed across three independent cryptographic primitives:

DSTU GOST 34.311-95

256-bit national legacy digest for QTSP interoperability.

DSTU 7564:2014

“Kupyna-256” sponge-based national cryptographic hash.

FIPS 180-4 SHA-256

Universal cross-runtime content commitment H_{doc}.

STEP 02 · ELLIPTIC CURVE SIGNATURE OVER GF(2^257)

Polynomial-Basis Curve DSTU_PB_257 & RFC 6979 Determinism

We operate over the binary Galois field $\mathbb{F}_{2^{257}} = \mathbb{F}_2[t]/(t^{257} + t^{12} + 1)$ on the non-supersingular elliptic curve:

$$E(\mathbb{F}_{2^{257}}): \quad y^2 + xy = x^3 + ax^2 + b, \quad a = 0, \quad \deg(f(t)) = 257$$

Using **IETF RFC 6979** deterministic nonce derivation (eliminating RNG side-channel risks), the signer’s key produces the signature pair (r, s) which is bound into a keyless public verification digest:

$$V = \text{SHA-256}(r_{\text{hex}} \parallel :: \parallel s_{\text{hex}}) \in \{0,1\}^{256}$$

24-Bit Quantized Sub-Micron Perturbation ($|\delta_k| \leq 10^{-5}$)

How 216 bits of the cryptographic signature vector are injected into the 9 Sri Yantra triangles with zero symmetry drift.

MATHEMATICAL TRANSFORMATION · 9 × 24-BIT BLOCK QUANTIZATION

Bijective Mapping from Signature Digest V to Coordinate Shifts δ_k

The first 27 bytes (**216 bits**) of the binding digest V are partitioned into nine 24-bit unsigned big-endian integers $u_k = \mathrm{uint24}_{\mathrm{BE}}(V[3k:3k+3]) \in [0, 16,777,215]$ and mapped into bounded micro-perturbations:

$$\delta_{k+1} = \mathrm{round}_9\left(\left(\frac{2 \cdot u_k}{16,777,215} - 1\right) \times 10^{-5}\right), \quad k = 0, ..., 8, \quad \delta_{k+1} \in [-10^{-5}, +10^{-5}]$$

BILATERAL SYMMETRY INVARIANT · ZERO VISUAL DISTORTION

Exact Counter-Symmetric Vertex Modulation

To preserve the sacred bilateral symmetry axis $x = C_x = 150.0$, each triangle T_k is modulated counter-symmetrically:

$$\mathrm{lx}'_k = \mathrm{round}_9(\mathrm{lx}_k + \delta_k), \quad \mathrm{rx}'_k = \mathrm{round}_9(2C_x - \mathrm{lx}'_k), \quad \mathrm{by}'_k = \mathrm{round}_9(\mathrm{by}_k + 0.5\delta_k), \quad \mathrm{ay}'_k = \mathrm{ay}_k$$

HUMAN EYE IMPERCEPTIBILITY

Max shift is 10^{-5} units (≈ 0.003 pt $\approx 1\mu\mathrm{m}$), which is **30× smaller** than a 300 DPI laser printer dot.

COMBINATORIAL ENTROPY

$(2^{24})^9 = 2^{216} \approx 1.05 \times 10^{65}$ distinct geometric states bound via SHA-256 commitment.

Detecting a 0.001 Coordinate Shift or 1-Byte Text Edit

Any modification to document text or vector geometry immediately breaks the 9-triangle verification matrix.

STATE A · AUTHENTIC SEAL

✅ All 9 Layers Verified

Marma_Crypto_Whitepaper_EN.pdf

- **Content SHA-256:** Matches embedded seal
- **DSTU 4145-2002 (r, s):** Valid over $\mathbb{F}_{2^{257}}$
- **Symmetry Error:** $\Delta_{\text{sym}} = 0.000000000000$
- **Coord Drift:** $\Delta_{\text{coord}} = 0.000000000000$

STATE B · SIMULATED ATTACK (+0.001 ON U2)

❌ Tamper Localized at Layer U2

Attack: U2.by + 0.001 | U2.lx - 0.001

- **Public Seal Digest:** Hash mismatch detected
- **Layer U2 Drift:** $\Delta_{\text{coord}} = 0.001000000000$
- **Symmetry Breach:** $\Delta_{\text{sym}} = 0.001000000000$
- **Forensic Verdict: REJECTED (FORGERY)**

EXCERPT FROM LIVE 9-TRIANGLE FORENSIC VERIFICATION MATRIX

Pinpointing Sub-Micron Tampering Across All 36 Coordinates

T_k	lx'_k	rx'_k	by'_k	δ_k	Δ_{coord}	Status
D1	53.65669103	246.34330897	123.20507847	-4.57e-6	0.000000000	OK
U1	52.98401843	247.01598157	174.24660931	+7.40e-6	0.000000000	OK
U3	98.71822977	201.28177023	220.03828779	-3.36e-6	0.000000000	OK
U2	78.26367665	221.73532335	197.92415508	-3.33e-6	0.001000000	TAMPER!
D3	90.48569103	209.51430897	78.10499115	-1.26e-6	0.000000000	OK
D2	80.98384056	219.01615944	103.12198754	-7.83e-6	0.000000000	OK

Why Marma-Crypto Outperforms Classical Signatures & Watermarks

Side-by-side comparison against RSA-3072, ECDSA P-256 / Ed25519, Detached CAdES, and Raster Steganography.

MULTI-DIMENSIONAL BENCHMARK MATRIX · CRYPTOGRAPHY × GEOMETRY × EVIDENTIARY LAW

Comparative Architecture Matrix

Criterion / Layer	RSA-3072 (PKCS#1 v2.2)	ECDSA P-256 / Ed25519	Raster / QR Watermark	🔱 Marma-Crypto v1.2.1 (DSTU 4145 × Sri Yantra)
Algebraic Group	Integer Ring $\mathbb{Z}_n^*$	Prime Field $\mathbb{F}_p$	None (Pixels)	**Binary Field $\mathbb{F}_{2^{257}}$**
Key / Sig Size (128-bit)	3072 / 3072 bit	256 / 512 bit	N/A	**257 / 512 bit (Carry-less)**
Digest Primitives	Single (SHA-256)	Single (SHA-256)	None / CRC	**Triple (GOST+Kupyna+SHA256)**
Nonce RNG Safety	Deterministic / PSS	Vulnerable if RNG weak	N/A	**Deterministic RFC 6979**
Visible Vector Binding	❌ None (Hidden ASN.1)	❌ None (Hidden ASN.1)	⚠ Decorative only	✅ **54 Marmas ($\lvert\delta_k\rvert \le 10^{-5}$)**
Tamper Localization	❌ Binary Pass/Fail	❌ Binary Pass/Fail	⚠ Visual degradation	✅ **Exact Layer $T_1..T_9$ (36 coords)**
Cross-Format Survival	❌ Lost on export	❌ Lost on export	❌ Breaks on scaling	✅ **Native .pdf, .docx, .svg**
Archival Legal Force	Requires PAdES	Requires PAdES	❌ Zero legal force	✅ **PAdES-B-LTA + TSP + OTS**

ADVANTAGE 01 · ALGEBRA

GF(2^257) Efficiency

Polynomial-basis arithmetic over $\mathbb{F}_2[t]/(t^{257}+t^{12}+1)$ uses carry-less XOR addition and provides cryptographic diversity beyond prime curves.

ADVANTAGE 02 · RESILIENCE

Dual-Channel Survival

Even if ASN.1 metadata is stripped by a document gateway, the 9-triangle vector coordinates retain the 216-bit signature commitment.

ADVANTAGE 03 · FORENSICS

Sub-Micron Pinpointing

Instead of an opaque “Invalid Signature” error, forensic auditors can localize the exact modified triangle and coordinate drift Δ_{coord}.

Statutory Legal Validity & 25-Year Archival Non-Repudiation

Engineered by a practicing litigation & digital economy attorney to satisfy strict courtroom evidentiary standards.

ETSI EN 319 142-1 · ISO 32000-2

PAdES-B-LTA Archival Container

Embeds full X.509 certificate chains, OCSP responses, and CRL revocation data directly inside the PDF Document Security Store (`/DSS`), guaranteeing offline verifiability for **25+ years** even after CA certificate expiration.

IETF RFC 3161 · ACCREDITED QTSP

Qualified Time-Stamp Token (TSP)

Cryptographically anchors the exact second of signing via accredited Qualified Trust Service Providers (**KNEDP** / **CZO** / **monobank** / **CA Diia**), preventing backdating or key-compromise repudiation.

UKRAINIAN LAW NO. 2155-VIII & eIDAS

Statutory Equivalence to Handwritten Seal

Fully compliant with the Law of Ukraine **“On Electronic Identification and Electronic Trust Services”** and EU **eIDAS Regulation (EU) No 910/2014** (`czo.gov.ua/verify` verified).

DECENTRALIZED PRIOR-ART ANCHORING

OpenTimestamps Bitcoin Proofs (.ots)

Every release manifest and whitepaper SHA-256 digest is anchored into the Bitcoin blockchain via **OpenTimestamps**, providing trustless global proof of existence.

SELF-REFERENTIAL CRYPTOGRAPHIC PROOF · LIVE IN V1.2.1

The Whitepapers Are Sealed Using the Exact Protocol They Specify

Signer: Yaroslav Volodymyrovych Baienko · CA JSC Universal Bank (monobank) · TSP: 2026-10-17 12:47 UTC

Both the English (`Marma_Crypto_Whitepaper_EN.pdf`) and Ukrainian (`Marma_Crypto_Whitepaper_UA.pdf`) editions carry live PAdES-B-LTA signatures + embedded 9-triangle Sri Yantra seals verified by a 9-suite automated test harness.

Zero-Trust Online Verifier & Interactive 9-Triangle Inspector

Verify privileged legal contracts (.pdf, .docx, .svg) directly in your browser — zero bytes ever leave your device.

ATTORNEY-CLIENT PRIVILEGE BY DESIGN · ZERO SERVER UPLOADS

Pure Client-Side Cryptographic & Geometric Engine

Law firms, banks, and courts cannot upload confidential M&A contracts or litigation filings to third-party cloud verifiers. Our web verifier executes **100% locally inside your browser sandbox**:

MULTI-FORMAT PARSER

Drag-and-drop any signed `.pdf`, `.docx` (OOXML custom properties + paragraphs + tables), or `.svg` file for instant verification.

INTERACTIVE GEOMETRY

Hover over any of the 9 Sri Yantra triangles (`D1..D5`, `U1..U4`) to inspect its exact vertices and sub-micron perturbation δ_k.

LIVE ATTACK SIMULATOR

One-click **“⚡ Simulate Attack (0.001)”** demonstrates real-time tamper detection and reset right in the browser.

PRINTABLE COURT PROTOCOL

Generate and print a formal bilingual Cryptographic Verification Certificate for court exhibits with one click.

TRY IT LIVE RIGHT NOW IN YOUR BROWSER

https://yaroslavbaienko.github.io/marma-crypto-protocol/

Pre-loaded with the official English & Ukrainian Whitepapers and Canonical Sri Yantra SVG so you can inspect the 54 triple marmas and test the 1-micron attack simulator immediately.

1-Command Installation & GNOME Desktop Integration

Production-tested packages published in our official GPG-signed Debian/Ubuntu APT repository.

OFFICIAL APT REPOSITORY · https://yaroslavbaienko.github.io/legaltech/

Install Both Sovereign LegalTech Suites in Seconds

```
# 1. Trust the repository GPG key (RSA-4096)
curl -fsSL https://yaroslavbaienko.github.io/legaltech/public.gpg | sudo gpg --dearmor -o /etc/apt/keyrings/legaltech.gpg

# 2. Register the LegalTech APT source
echo "deb [signed-by=/etc/apt/keyrings/legaltech.gpg] https://yaroslavbaienko.github.io/legaltech stable main" | sudo tee /etc/apt/sources.list.d/legaltech.list

# 3. Install qes-tools & marma-crypto-watermark
sudo apt update && sudo apt install qes-tools marma-crypto-watermark
```

PACKAGE 1 · AMD64 (v1.0.26)

🔐 qes-tools

- **Standards:** DSTU 4145-2002, GOST 34.311-95, Kupyna-256, PAdES-B-LTA, CAdES-X-Long, ASiC-E, RFC 3161 TSP.
- **Hardware Tokens:** DepositSign, IIT Almaz-1K, Crystal-1K, SecureToken-337.
- **Tools:** `qes-sign`, `qes-verify`, `qes-pdf-court`, `qes-ocr`, plus 18 GNOME Files (Nautilus) actions.

PACKAGE 2 · ALL (v1.2.1-1)

🔱 marma-crypto-watermark

- **Formats:** PDF, DOCX, and SVG dual-layer sealing (`--mode both | watermark | seal`).
- **Desktop GUI:** GTK4 / Libadwaita key selector (`marma-dialog`) & 15-min RAM session cache.
- **CLI:** `marma-watermark sign | verify | protocol | selftest` + Nautilus right-click integration.

Enterprise Integration & Forensic Legal Advisory

Partner with us to deploy mathematical document non-repudiation across banking, legal, M&A, and Web3 workflows.

PACKAGE 01 · AUTOMATED PIPELINES & API

🏛️ Enterprise Document Sealing CI/CD

Turnkey integration of **Marma-Crypto + Qualified Electronic Signatures (DSTU 4145-2002 / eIDAS PAdES-B-LTA + RFC 3161 TSP)** into high-volume banking, law firm, M&A data room, and RWA/Web3 document pipelines via headless Linux CLI, Docker, or on-premise REST/gRPC API.

PACKAGE 02 · INSTITUTIONAL CRYPTOGRAPHIC IDENTITY

🔱 Custom Corporate Geometric Seal

Dedicated organizational geometric-cryptographic seal bound to your corporate **HSM / PKCS11 / QES certificate**, custom white-label browser verification portal, and brand-aligned vector templates for high-value contracts, share certificates, and board resolutions.

PACKAGE 03 · LITIGATION & INTERNATIONAL ARBITRATION

⚖️ Forensic Document Authenticity & Expert Opinion

Independent cryptographic and geometric forensic examination of disputed electronic evidence (`.pdf`, `.docx`, `.svg`, `.p7s`), sub-micron tamper localization across all 36 triangle coordinates, and formal legal/technical expert reports for courts and international arbitration.

CONNECT WITH THE AUTHOR & EXPLORE THE ECOSYSTEM

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🌐 Verifier: yaroslavbaienko.github.io/marma-crypto-protocol

📄 Zenodo DOI: doi.org/10.5281/zenodo.23436152

🐧 APT Repo: yaroslavbaienko.github.io/legaltech

𝕏 / Product Hunt: @baienko · p/marma-crypto-protocol